T10: FORMAS CANONICAS

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1. Simplificar: $\overline{\overline{a+b}+\bar{c}}$

(★) $\overline{\bar{a}\cdot\bar{b}+\bar{c}}$

(★) $\bar{\bar{a}}+\bar{\bar{b}}\cdot\bar{\bar{c}}$

(★) $\boxed{a+b\cdot c}$

2. Simplificar: $f(x,y,z)=x+\bar{x}y+xy\bar{z}+xz+x\bar{z}$

$x+\bar{x}y+xy\bar{z}+xz+x\bar{z}$

(★) $x+y\cdot(\bar{x}+x\bar{z})+x\cdot(z+\bar{z})$

(★) $x+y\cdot((\bar{x}+x)(\bar{x}+\bar{z})+x(1)$

(★) $x+y\cdot(1)(\bar{x}+\bar{z})+x$

(★) $x+y(\bar{x}+\bar{z})+x$

(★) $x+y\bar{x}+y\bar{z}+x$

(★) $x+\bar{x}y+y\bar{z}$

(★) $x\cdot 1+\bar{x}y+y\bar{z}$

(★) $x\cdot(y+\bar{y})+\bar{x}y+y\bar{z}$

(★) $xy+x\bar{y}+\bar{x}y+y\bar{z}$

(★) $y\cdot(x+\bar{x})+x\bar{y}+y\bar{z}$

(★) $y\cdot(1)+x\bar{y}+y\bar{z}$

(★) $y+x\bar{y}+y\bar{z}$

(★) $y(1+\bar{z})+x\bar{y}$

(★) $y(1)+x\bar{y}$

(★) $y+x\bar{y}$

3. Simplificar: $f(a,b)=\overline{(a+b)}(\overline{\bar{a}+b})$

(★) $\overline{(a+b)}(\overline{\bar{a}+b})$

(★) $(\bar{a}\cdot\bar{b})(\bar{\bar{a}}\cdot\bar{b})$

(★) $(\bar{a}\cdot\bar{b})(a\cdot\bar{b})$

(★) $\bar{b}(\bar{a}\cdot a)$

(★) $\bar{b}(0)$

(★) 0

4. Evalúe utilizando una tabla de verdad: $f(a,b,c) = \overline{(a \oplus c)} + \overline{a}\,\overline{b} + \overline{bc}$

Nro	a	b	c	$\bar{a}$	$\bar{b}$	$a\oplus c$	$\overline{a\oplus c}$	$\bar{a}\bar{b}$	bc	$\overline{bc}$	F(a,b,c)
0	0	0	0	1	1	0	1	1	0	1	1
1	0	0	1	1	1	1	0	1	0	1	1
2	0	1	0	1	0	0	1	0	0	1	1
3	0	1	1	1	0	1	0	0	1	0	0
4	1	0	0	0	1	1	0	0	0	1	1
5	1	0	1	0	1	0	1	0	0	1	1
6	1	1	0	0	0	1	0	0	0	1	1
7	1	1	1	0	0	0	1	0	1	0	1

5. Expresar en su forma algebraica: $h(a,b,c,d) = \sum(0,2,5,7,11,12,13,15)$

Nro	a	b	c	d	h	
0	0	0	0	0	1	→ $\bar{a}\bar{b}\bar{c}\bar{d}$
1	0	0	0	1	0	
2	0	0	1	0	1	→ $\bar{a}\bar{b}c\bar{d}$
3	0	0	1	1	0	
4	0	1	0	0	0	
5	0	1	0	1	1	→ $\bar{a}b\bar{c}d$
6	0	1	1	0	0	
7	0	1	1	1	1	→ $\bar{a}bcd$
8	1	0	0	0	0	
9	1	0	0	1	0	
10	1	0	1	0	0	
11	1	0	1	1	1	→ $a\bar{b}cd$
12	1	1	0	0	1	→ $ab\bar{c}\bar{d}$
13	1	1	0	1	1	→ $ab\bar{c}d$
14	1	1	1	0	0	
15	1	1	1	1	1	→ $abcd$

$$\boxed{\bar{a}\bar{b}\bar{c}d + \bar{a}\bar{b}c\bar{d} + \bar{a}b\bar{c}d + \bar{a}bcd + a\bar{b}cd + ab\bar{c}\bar{d} + ab\bar{c}d + abcd}$$

6. Dibuje la tabla de verdad para la función k: $k(x,y,z,w)=\Pi(1,2,4,8)+d(0,9)$

Nro	x	y	z	w	k(x y z w)
0	0	0	0	0	X
1	0	0	0	1	0
2	0	0	1	0	0
3	0	0	1	1	1
4	0	1	0	0	0
5	0	1	0	1	1
6	0	1	1	0	1
7	0	1	1	1	1
8	1	0	0	0	0
9	1	0	0	1	X
10	1	0	1	0	1
11	1	0	1	1	1
12	1	1	0	0	1
13	1	1	0	1	1
14	1	1	1	0	1
15	1	1	1	1	1

7. Obtener la expresión booleana en SdP especificada por la siguiente tabla:

Nro	x	y	z	F(x,y,z)	
0	0	0	0	0	
1	0	0	1	0	
2	0	1	0	0	
3	0	1	1	0	
4	1	0	0	1	$\rightarrow x\bar{y}\bar{z}$
5	1	0	1	0	
6	1	1	0	1	$\rightarrow xy\bar{z}$
7	1	1	1	0	

$x\bar{y}\bar{z} + xy\bar{z}$

8. Obtener la expresión booleana en PdS especificada por la siguiente tabla:

Nro	x	y	z	F(x,y,z)	
0	0	0	0	1	
1	0	0	1	0	$\rightarrow (x+y+\bar{z})$
2	0	1	0	1	
3	0	1	1	1	
4	1	0	0	1	
5	1	0	1	0	$\rightarrow (\bar{x}+y+\bar{z})$
6	1	1	0	1	
7	1	1	1	1	

$(x+y+\bar{z})(\bar{x}+y+\bar{z})$

9. Representar en una tabla de verdad los todos los min términos y los Max términos en su notación algebraica para una función de 3 variables.

Nro	a	b	c	F(a,b,c)		
0	0	0	0	0		$a+b+c$
1	0	0	1	0		$a+b+\bar{c}$
2	0	1	0	1	→ $\bar{a}b\bar{c}$	
3	0	1	1	0		$a+\bar{b}+\bar{c}$
4	1	0	0	1	→ $a\bar{b}\bar{c}$	
5	1	0	1	0		$\bar{a}+b+\bar{c}$
6	1	1	0	1	→ $ab\bar{c}$	
7	1	1	1	1	→ abc	

f(a,b,c) Σmin (2,4,6,7)

Πmax (0,1,3,5)

10. Sea la tabla con los números del o 15 que representan las 16 combinaciones para 4 variables, y sea *F* la función correspondiente, obtener las dos formas canónicas para *F*, escribirlas algebraicamente y como la suma/producto de sus términos:

Nro	F	a	b	c	d		
0	x	0	0	0	0		
1	1	0	0	0	1	$\bar{a}\bar{b}\bar{c}d$	
2	x	0	0	1	0		
3	0	0	0	1	1		$a+b+\bar{c}+\bar{d}$
4	0	0	1	0	0		$a+\bar{b}+c+d$
5	0	0	1	0	1		$a+\bar{b}+c+\bar{d}$
6	0	0	1	1	0		$a+\bar{b}+\bar{c}+d$
7	1	0	1	1	1	$\bar{a}bcd$	
8	0	1	0	0	0		$\bar{a}+b+c+d$
9	1	1	0	0	1	$a\bar{b}\bar{c}d$	
10	0	1	0	1	0		$\bar{a}+b+\bar{c}+d$
11	x	1	0	1	1		
12	0	1	1	0	0		$\bar{a}+\bar{b}+c+d$
13	1	1	1	0	1	$ab\bar{c}d$	
14	1	1	1	1	0	$abc\bar{d}$	
15	1	1	1	1	1	$abcd$	

Σmin(1,7,9,13,14,15) + d (0,2,11)

Π Máx(3,4,5,6,8,10,12) + d(0,2,11)

11. Sea $G = \sum min\,(7,11,13,14,15) + d(\,0,1,9)$, escribirla en la forma Producto de sumas.

Suma

Nro	a	b	c	d	G	
0	0	0	0	0	X	
1	0	0	0	1	X	
2	0	0	1	0	0	→ $a+b+\bar{c}+d$
3	0	0	1	1	0	→ $a+b+\bar{c}+\bar{d}$
4	0	1	0	0	0	→ $a+\bar{b}+c+d$
5	0	1	0	1	0	→ $a+\bar{b}+c+\bar{d}$
6	0	1	1	0	0	→ $a+\bar{b}+\bar{c}+d$
7	0	1	1	1	1	
8	1	0	0	0	0	→ $\bar{a}+b+c+d$
9	1	0	0	1	X	
10	1	0	1	0	0	→ $\bar{a}+b+\bar{c}+d$
11	1	0	1	1	1	
12	1	1	0	0	0	→ $\bar{a}+\bar{b}+c+d$
13	1	1	0	1	1	
14	1	1	1	0	1	
15	1	1	1	1	1	

$(a+b+\bar{c}+d)(a+b+\bar{c}+\bar{d})(a+\bar{b}+c+d)(a+\bar{b}+c+\bar{d})(a+\bar{b}+\bar{c}+d)(\bar{a}+b+c+d)$
$(\bar{a}+b+\bar{c}+d)(\bar{a}+\bar{b}+c+d)$

12. Sean las funciones:

$f = \prod(0,1,2,6,7,13,14,15)\ y\, h = \sum \min(2,5,7,10,14,15)\,; hallar\ I = \overline{f * h} + f \oplus \mathrm{h}$

Dónde:

a	b	$a \oplus \mathrm{b} = \mathrm{a\bar{b}} + \mathrm{\bar{a}b}$
0	0	0
0	1	1
1	0	1
1	1	0

Nro	F	h	$\overline{F\cdot h}$	$F\oplus h$	$I = \overline{F\cdot h} + F\oplus h$
0	0	0	1	0	1
1	0	0	1	0	1
2	0	1	1	1	1
3	1	0	1	1	1
4	1	0	1	1	1
5	1	1	0	0	0
6	0	0	1	0	1
7	0	1	1	1	1
8	1	0	1	1	1
9	1	0	1	1	1
10	1	1	0	0	0
11	1	0	1	1	1
12	1	0	1	1	1
13	0	0	1	0	1
14	0	1	1	1	1
15	0	1	1	1	1

13. Sea la función: $f = a(cd + b)$, obtener sus dos formas canónicas por el método algebraico.

★ $f = a(cd+b)$

$acd + ab$

$acd \cdot 1 + ab \cdot 1 \cdot 1$

$acd \cdot (b+\bar{b}) + ab \cdot (c+\bar{c})(d+\bar{d})$

$abcd + a\bar{b}cd + (abc + ab\bar{c})(d+\bar{d})$

$abcd + a\bar{b}cd + abcd + ab\bar{c}d + abc\bar{d} + ab\bar{c}\bar{d}$

(★) $abcd + a\bar{b}cd + ab\bar{c}d + abc\bar{d} + ab\bar{c}\bar{d}$

PS

$f = a(cd+b)$

★ $acd + ba \rightarrow (acd)+(ba)$

★ $(acd+a)(acd+b)$

★ $(a+a)(a+c)(a+d)(b+a)(b+c)(b+d)$

★ $(a+0+0+0)(a+c+0+0)(a+d+0+0)(b+a+0+0)(b+c+0+0)(b+d+0+0)$

★ $(a+b\bar{b}+c\bar{c}+d\bar{d})(a+c+b\bar{b}+d\bar{d})(a+d+b\bar{b}+c\bar{c})(b+a+c\bar{c}+d\bar{d})(b+c+a\bar{a}+d\bar{d})(b+d+a\bar{a}+c\bar{c})$

– ★ $(a+b+c\bar{c}+d\bar{d})(a+\bar{b}+c\bar{c}+d\bar{d})(a+c+b+d\bar{d})(a+c+\bar{b}+d\bar{d})(\bar{a}+d+b+c\bar{c})(a+d+\bar{b}+c\bar{c})$
$(b+a+c+d\bar{d})(b+a+\bar{c}+d\bar{d})(b+c+a+d\bar{d})(b+c+\bar{a}+d\bar{d})(b+d+a+c\bar{c})(b+d+\bar{a}+c\bar{c})$

★ $(a+b+c+d\bar{d})(a+b+\bar{c}+d\bar{d})(a+\bar{b}+c+d\bar{d})(a+\bar{b}+\bar{c}+d\bar{d})$

★ $(a+b+c+d)(a+b+c+\bar{d})(a+b+\bar{c}+d)(a+b+\bar{c}+\bar{d})(a+\bar{b}+c+d)(a+\bar{b}+c+\bar{d})(a+\bar{b}+\bar{c}+d)$
$(a+\bar{b}+\bar{c}+\bar{d})(a+c+b+d)(a+c+b+\bar{d})(a+\bar{b}+c+d)(a+\bar{b}+c+\bar{d})(a+b+c+d)(a+b+\bar{c}+d)$
$(a+\bar{b}+c+d)(a+\bar{b}+\bar{c}+d)(a+b+c+d)(a+b+c+\bar{d})(a+b+\bar{c}+d)(a+b+\bar{c}+\bar{d})(a+b+c+d)$
$(a+b+c+\bar{d})(\bar{a}+b+c+d)(\bar{a}+b+c+\bar{d})(a+b+c+d)(a+b+\bar{c}+d)(\bar{a}+b+c+d)(\bar{a}+b+\bar{c}+d)$

★ $(a+b+c+d)(a+b+c+\bar{d})(a+b+\bar{c}+d)(a+b+\bar{c}+\bar{d})(a+\bar{b}+c+d)(a+\bar{b}+c+\bar{d})(a+\bar{b}+\bar{c}+d)$
$(a+\bar{b}+\bar{c}+\bar{d})(\bar{a}+b+c+d)(\bar{a}+b+c+\bar{d})(\bar{a}+b+\bar{c}+d)$

14. Sea $g = \bar{x}y + \bar{w}xz$; obtener sus dos formas canónicas por el método algebraico.

SdP:

$$\bar{x}y + \bar{w}xz$$

$$\bar{x}y \cdot 1 \cdot 1 + \bar{w}xz \cdot 1$$

$$\bar{x}y \cdot (w+\bar{w})(z+\bar{z}) + \bar{w}xz\,(y+\bar{y})$$

$$(w\bar{x}y + \bar{w}\bar{x}y)(z+\bar{z}) + \bar{w}xyz + \bar{w}x\bar{y}z$$

$$w\bar{x}yz + \bar{w}\bar{x}yz + w\bar{x}y\bar{z} + \bar{w}\bar{x}y\bar{z} + \bar{w}xyz + \bar{w}x\bar{y}z$$

PdS

$$(\bar{x}y) + (\bar{w}xz)$$

$$(\bar{x}y+\bar{w})(\bar{x}y+x)(\bar{x}y+z)$$

$$(\bar{w}+\bar{x})(\bar{w}+y)(x+\bar{x})(x+y)(z+\bar{x})(z+y)$$

$$(\bar{w}+\bar{x}+0+0)(\bar{w}+y+0+0)(1)(x+y+0+0)(\bar{x}+z+0+0)(y+z+0+0)$$

$$(\bar{w}+\bar{x}+y\bar{y}+z\bar{z})(\bar{w}+y+x\bar{x}+z\bar{z})(x+y+w\bar{w}+z\bar{z})(\bar{x}+z+y\bar{y}+w\bar{w})(y+z+x\bar{x}+w\bar{w})$$

$$(\bar{w}+\bar{x}+y+z\bar{z})(\bar{w}+\bar{x}+\bar{y}+z\bar{z})(\bar{w}+y+x+z\bar{z})(\bar{w}+y+\bar{x}+z\bar{z})(x+y+w+z\bar{z})(x+y+\bar{w}+z\bar{z})$$
$$(\bar{x}+z+y+w\bar{w})(\bar{x}+z+\bar{y}+w\bar{w})(y+z+x+w\bar{w})(y+z+\bar{x}+w\bar{w})$$

$$(\bar{w}+\bar{x}+y+z)(\bar{w}+\bar{x}+y+\bar{z})(\bar{w}+\bar{x}+\bar{y}+z)(\bar{w}+\bar{x}+\bar{y}+\bar{z})(\bar{w}+x+y+z)(\bar{w}+x+y+\bar{z})(\bar{w}+\bar{x}+y+z)$$
$$(\bar{w}+\bar{x}+y+\bar{z})(w+x+y+z)(w+x+y+\bar{z})(\bar{w}+x+y+z)(\bar{w}+x+y+\bar{z})(w+\bar{x}+y+z)(\bar{w}+\bar{x}+y+z)$$
$$(w+\bar{x}+\bar{y}+z)(\bar{w}+\bar{x}+\bar{y}+z)(w+x+y+z)(\bar{w}+x+y+z)(w+\bar{x}+y+z)(\bar{w}+\bar{x}+y+z)$$

$$(\bar{w}+\bar{x}+y+z)(\bar{w}+\bar{x}+y+\bar{z})(\bar{w}+\bar{x}+\bar{y}+z)(\bar{w}+\bar{x}+\bar{y}+\bar{z})(\bar{w}+x+y+z)(\bar{w}+x+y+\bar{z})(w+x+y+z)$$
$$(w+x+y+\bar{z})(w+\bar{x}+y+z)(w+\bar{x}+\bar{y}+z)$$

15. Sea la función *f* expresada en P.d.S., obtener la forma canónica S.d.P

$$f(A,B,C,D) = \Pi(0,2,4,6,8,10,12,14)$$

Nº	A	B	C	D	F(ABCD)		
0	0	0	0	0	0		
1	0	0	0	1	1	→	$\bar{A}\bar{B}\bar{C}D$
2	0	0	1	0	0		
3	0	0	1	1	1	→	$\bar{A}\bar{B}CD$
4	0	1	0	0	0		
5	0	1	0	1	1	→	$\bar{A}B\bar{C}D$
6	0	1	1	0	0		
7	0	1	1	1	1	→	$\bar{A}BCD$
8	1	0	0	0	0		
9	1	0	0	1	1	→	$A\bar{B}\bar{C}D$
10	1	0	1	0	0		
11	1	0	1	1	1	→	$A\bar{B}CD$
12	1	1	0	0	0		
13	1	1	0	1	1	→	$AB\bar{C}D$
14	1	1	1	0	0		
15	1	1	1	1	1	→	$ABCD$

$(\bar{A}\bar{B}\bar{C}D) + \bar{A}\bar{B}CD + \bar{A}B\bar{C}D + \bar{A}BCD + A\bar{B}\bar{C}D + A\bar{B}CD + AB\bar{C}D + ABCD$

16. Obtener mediante el método algebraico las dos formas canónicas de la función:

K = $\overline{\overline{a+b} + \bar{c}}$

★ $\overline{\bar{a} \cdot \bar{b} + \bar{c}}$

$\bar{\bar{a}} + \bar{\bar{b}} \cdot \bar{\bar{c}} \rightarrow a + b \cdot c \rightarrow ac + bc$

$ac \cdot 1 + bc \cdot 1$

$ac \cdot (b+\bar{b}) + bc \cdot (a+\bar{a}) \rightarrow abc + a\bar{b}c + abc + \bar{a}bc = abc + a\bar{b}c + \bar{a}bc$

PdS

★ $ac + bc$

$(ac+b)(ac+c)$

$(a+b)(c+b)(a+c)(c+c)$

$(a+b+0)(b+c+0)(a+c+0)(c+0+0)$

$(a+b+c\bar{c})(b+c+a\bar{a})(a+c+b\bar{b})(c+a\bar{a}+b\bar{b})$

$(a+b+c)(a+b+\bar{c})(a+b+c)(\bar{a}+b+c)(a+b+c)(a+\bar{b}+c)(a+b\bar{b}+c)(\bar{a}+b\bar{b}+c)$

$(a+b+c)(a+b+\bar{c})(a+b+c)(\bar{a}+b+c)(a+b+c)(a+\bar{b}+c)(a+b+c)(a+\bar{b}+c)(\bar{a}+b+c)(\bar{a}+\bar{b}+c)$

$(a+b+c)(a+b+\bar{c})(\bar{a}+b+c)(a+\bar{b}+c)(\bar{a}+\bar{b}+c)$